https://www.istockphoto.com/photos/people-in-movie-theater

Do Movie Sequels Perform Better Than Originals?

Think about some of the biggest movie franchises you've seen in theaters.

Star Wars, Marvel, Fast & Furious, and many others.

Sequels are everywhere in modern cinema — but it got me thinking about a simple question:

Do sequels actually perform better than the original film that started the franchise?

Curious to find out, I explored this question using **Python** and a movie dataset I found on **Kaggle.**

By analyzing thousands of films, I focused on two key questions:

- Do sequels typically generate more box office revenue than the original?
- Are there patterns in the data that help predict when a sequel will succeed?

Using my data analytics and machine learning skills, I analyzed the dataset to see what insights the numbers would reveal.

Here's what I found.

How I Defined a “Successful Sequel”

For this project, I used a very simple definition:

A successful sequel is a sequel that earns more box office revenue than the original film.

This avoids subjective measures like reviews or popularity and focuses purely on **financial performance** of the film.

The Data

This analysis uses **The Movies Dataset**, which contains metadata for roughly **45,000 films released on or before July 2017.** The dataset is derived from the **Full MovieLens Dataset** and includes a variety of movie attributes.

Key fields used in this project include:

- Box office revenue
- Production budget
- Audience ratings
- Vote counts
- Popularity scores
- Release dates
- Franchise (collection) membership

Movies were grouped into franchises (collections), such as *Star Wars, Fast & Furious,* and other multi-film series.

Within each franchise:

- The **first movie released** was treated as the **original film**
- Any movie released afterward was treated as a **sequel**

Each sequel was then compared directly to the original in that franchise.

What the Data Shows

Do Sequels Outperform Originals?

Here's what I found:

Average (Mean) Revenue

- **Mean original revenue:** $189,553,639
- **Mean sequel revenue:** $200,525,127

Median Revenue

- **Median original revenue:** $101,702,060
- **Median sequel revenue:** $83,080,890

Overall Outcome

- **Only about 42% of sequels earn more than the original film.**

What This Means

A small number of extremely successful sequels drive the **average earnings higher.**

However, when we look at the typical movie, most sequels actually earn **less than the original film that started the franchise**.

In other words, while blockbuster sequels get the most attention, **true sequel success is less common than it may appear.**

Visualizing Original vs Sequel Revenue

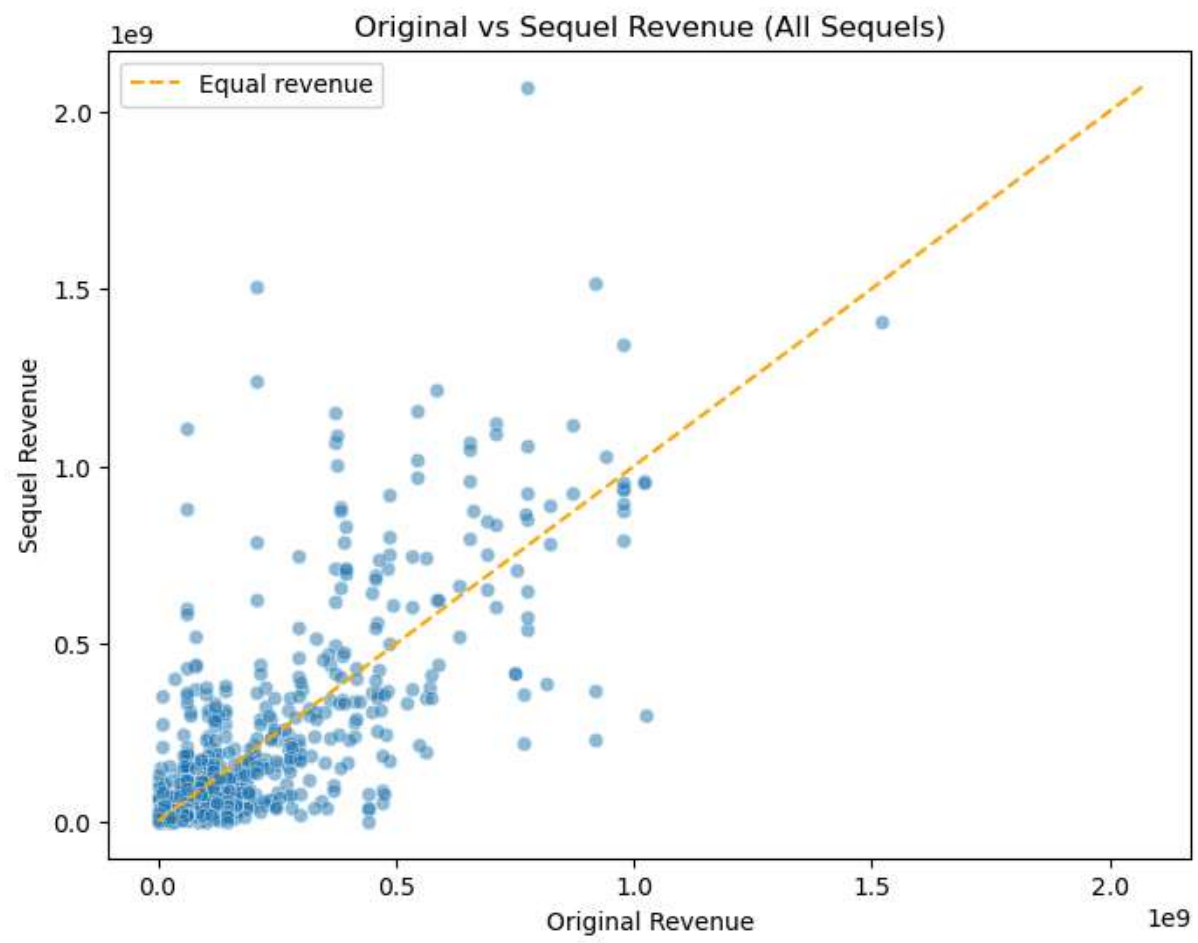
I created a **scatter plot comparing revenue between originals and sequels.**

Each point represents **one franchise comparison**:

- **X-axis:** Original revenue

- **Y-axis:** Sequel revenue
- **The orange line:** Equal revenue
- **The blue points**: Films

Most points fall **below the line**, showing that the **majority of sequels earn less than the original film.**



Takeaway: While blockbuster sequels do exist, they're the **exception — not the rule.**

The Sequels That Break the Pattern

Some sequels dramatically outperform their originals.

When examining the largest revenue increases, a clear pattern emerged.

The top performers come from **major global franchises**, including:

- *Star Wars*
- *Fast & Furious*
- *Dr No (James Bond franchise)*

These franchises benefit from:

- Massive brand recognition
- Global fan bases
- Strong marketing momentum

While most sequels struggle to beat the original, **a small number of mega-franchises dominate the top of the list.**

Collection Name	Original Title	Sequel Title	Installment Number	Original Revenue	Sequel Revenue	Revenue Increase	Years Since Original
The Fast and the Furious Collection	The Fast and the Furious	Furious 7	7	$207,283,925	$1,506,249,000	$1,298,965,075	13.774127
Star Wars Collection	Star Wars	Star Wars: The Force Awakens	7	$775,398,007	$2,068,224,000	$1,292,825,993	38.557153
James Bond Collection	Dr. No	Skyfall	25	$59,600,000	$1,108,561,000	$1,048,961,000	50.058864
The Fast and the Furious Collection	The Fast and the Furious	The Fate of the Furious	8	$207,283,925	$1,238,765,000	$1,031,481,075	15.805613
James Bond Collection	Dr. No	Spectre	26	$59,600,000	$880,674,600	$821,074,600	53.059548

Can We Predict a Successful Sequel?

Next, I trained a **machine learning model** to see if sequel success could be predicted.

The model used information from the original film, including:

- Audience rating
- Number of audience votes
- Budget
- Popularity score
- Runtime
- Time between releases
- Franchise installment number

Model Results

The model achieved:

- **Accuracy:** ~65%
- **ROC AUC:** ~0.66 - ROC AUC measures how well the model distinguishes **success from failure.**

The model performs **better than random guessing**, but it's not a crystal ball — strong predictive models often score **above 0.8.**

This isn't surprising.

Movie success depends on many unpredictable factors:

- Marketing
- Release timing
- Competition
- Cultural trends

Even with data, predicting which movies will succeed remains challenging.

What Factors Matter Most?

The model uncovered some interesting patterns about which factors influence sequel success - and which ones don't.

Factors Linked to Successful Sequels

Sequels tended to perform better when:

- The original film had higher ratings
- Audience engagement was strong
- The installment occurred later in an established franchise

In short: **Strong fan interest and franchise momentum matter more than anything else.**

Factors That Mattered Less

Some factors had little impact on sequel success, including:

- Bigger production budgets
- Popularity scores alone
- Longer runtimes

This challenges the common assumption that spending more guarantees a successful film.

Ratings vs Sequel Success

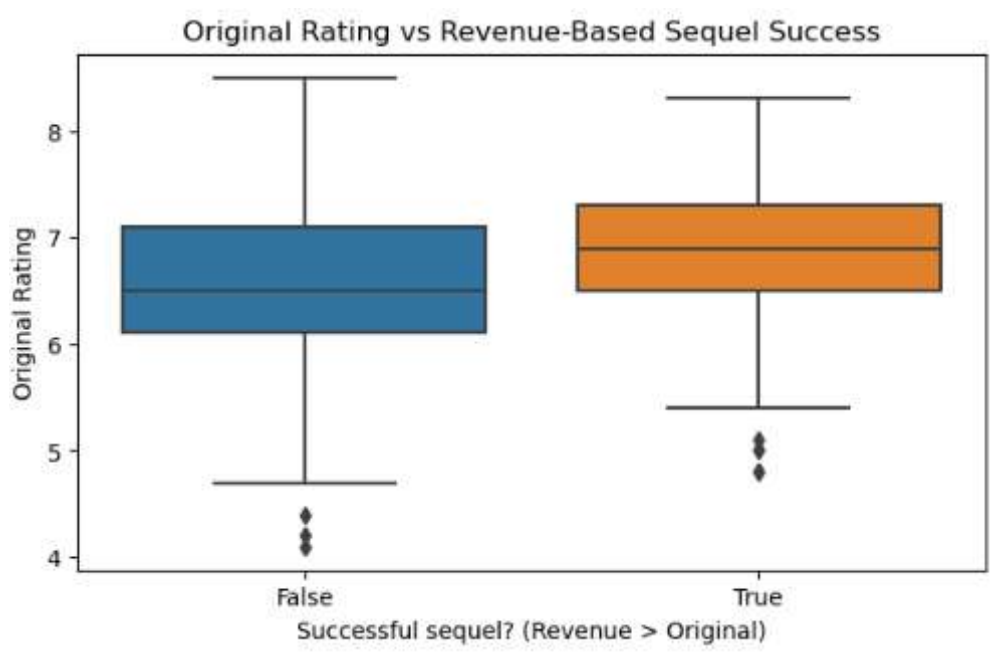
I also compared ratings of original films in two groups:

- Originals whose sequels made **less money (False)**
- Originals whose sequels made **more money (True)**

The results show that successful sequels tend to come from **slightly higher-rated original films.**

However, the difference isn't huge.

This suggests that **good original films help create successful franchises — but they don't guarantee it.**



What We Learned

This analysis revealed several key insights about sequel performance:

- **Most sequels underperform:** The majority earn less than the original.

- **Blockbusters skew the averages:** A few hugely successful franchises make sequels look more profitable than they typically are.
- **Audience engagement matters:** Strong ratings and fan interest are the best predictors of sequel success.
- **Big budgets aren't a guarantee:** Spending more doesn't automatically translate to success.
- **Prediction is possible, but uncertain:** Data can guide expectations, but predicting hits with certainty remains difficult.

Tools & Technologies

To conduct this analysis, I used the following tools:

Programming Language:

- Python

Libraries & Frameworks:

- Pandas
- NumPy
- Seaborn
- Matplotlib
- Scikit-learn

AI Assistance:

- ChatGPT (OpenAI)

These tools made it possible to clean, analyze, visualize, and model the data effectively.

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